
Math Training — M1 Cognitive Science

Session 1 — Linear Algebra I

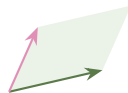
Vectors, span, dot product, change of basis, matrices

Advanced track · December 2026

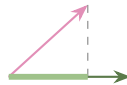
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Roadmap of the session

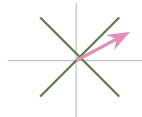
1. Vector, linear combination, **span**
2. Dot product $\longrightarrow$ **projection**
3. **Change of basis**: same point, new axes
4. **Matrix** = transformation of space



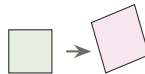
span



projection

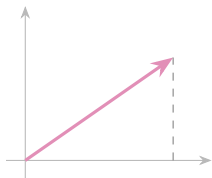


new axes

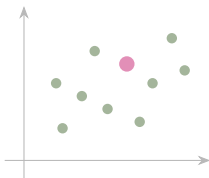


transformation

One object, three readings



arrow: length, angle



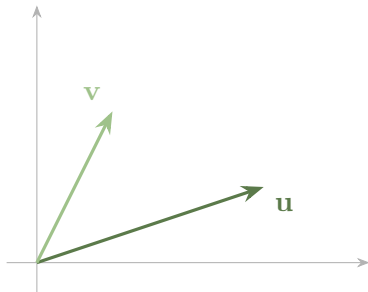
point: one row of a table **list:** how data arrives

$$\mathbf{p} = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix} \quad \text{one number per measure}$$

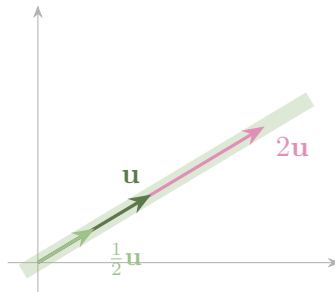
■ Definition

$\mathbf{v} \in \mathbb{R}^n$: an ordered list of n numbers. Two operations only — **add** slot by slot, **scale** $\lambda \mathbf{v} = (\lambda v_1, \dots, \lambda v_n)$.

Adding and scaling, drawn



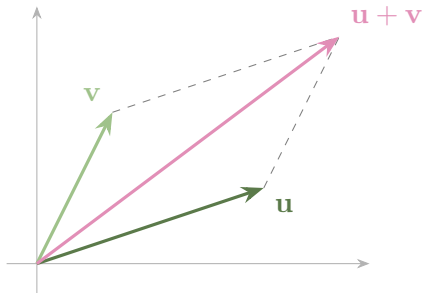
add: tip to tail



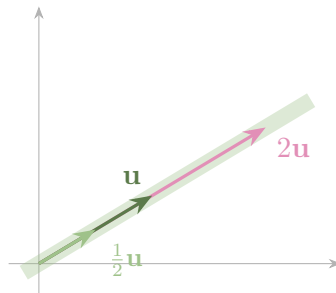
scale: never leaves the line

Left: the sum is the far corner of the parallelogram. Right: scaling slides the arrow along its own line — which is why one vector spans only a line.

Adding and scaling, drawn



add: tip to tail



scale: never leaves the line

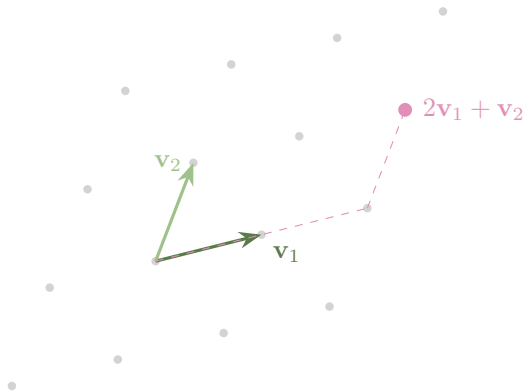
Left: the sum is the far corner of the parallelogram. Right: scaling slides the arrow along its own line — which is why one vector spans only a line.

■ Scale each one, then add them up

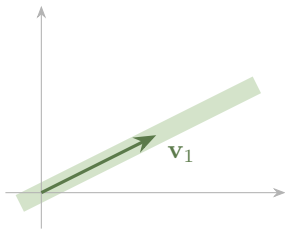
$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_k \mathbf{v}_k.$$

All of it is this one operation:

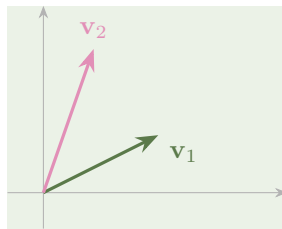
- span, dependence
- basis, coordinates
- matrix–vector product
- rank, eigenvectors
- principal components



Span: everything you can build



one vector $\Rightarrow$ a **line**

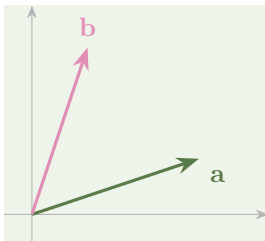


two independent $\Rightarrow$ a **plane**

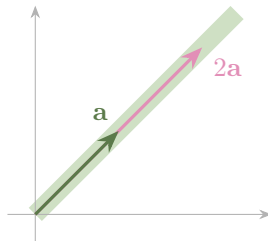
■ Definition

$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is the set of *all* their linear combinations: a line, a plane, or a flat of higher dimension — always through the origin.

Independent, or not



independent: the whole plane



dependent: one line only

■ Definition

Dependent $\iff \lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k = \mathbf{0}$ for some λ_i *not all zero*. Independent otherwise.

In $\mathbb{R}^n$, more than n vectors is always too many

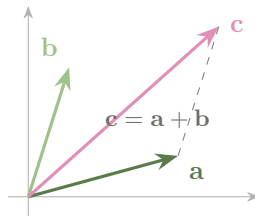
■ Useful fact

In $\mathbb{R}^n$: any $n + 1$ vectors are dependent,
and any n independent vectors span all of
 $\mathbb{R}^n$.

Three vectors in $\mathbb{R}^2$? Dependent.

No computation needed.

This is the fact *rank* will rest on next session.



■ Watch out

“They look different, so they are independent.”

$$(1, 2) \text{ \& } (2, 4) \quad (3, 3) \text{ \& } (1, 1)$$

Different-looking lists of numbers — each a multiple of the other.

The cure: ask for the vanishing combination explicitly. $2(1, 2) - 1(2, 4) = \mathbf{0}$

The running example: a data table

	RT	acc.	anx.
P_1	4	9	1
P_2	6	7	3
P_3	5	8	2
P_4	9	4	6
P_5	7	6	4

← a **row** = one participant

↑ a **column** = one measure, across everybody

■ Link to cognitive science

$P_2 = (6, 7, 3)$ is a point in $\mathbb{R}^3$; the accuracy column $(9, 7, 8, 4, 6)$ is a point in $\mathbb{R}^5$.

Same table, two different vector spaces — and which one you are in is decided by the question you are asking.

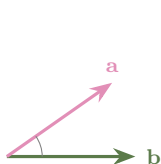
■ Practice — try it

1. $\mathbf{a} = (1, 2)$, $\mathbf{b} = (3, -1)$: compute $\mathbf{a} + \mathbf{b}$, $2\mathbf{a}$, $\mathbf{a} - \mathbf{b}$.
2. Independent? (i) $(1, 1)$ & $(2, 2)$; (ii) $(1, 0)$ & $(0, 1)$; (iii) $(2, -4)$ & $(-1, 2)$.
Give a vanishing combination for each dependent pair.
3. Describe $\text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$. Is $(1, 2, 1)$ in it?

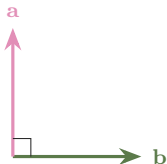
The dot product: one number, two formulas

■ Definition

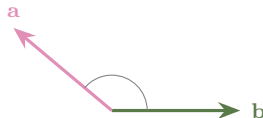
$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i = \|\mathbf{a}\| \|\mathbf{b}\| \cos \alpha, \quad \|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}.$$



$$\mathbf{a} \cdot \mathbf{b} > 0$$

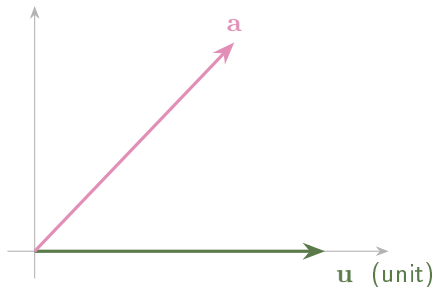


$$\mathbf{a} \cdot \mathbf{b} = 0$$



$$\mathbf{a} \cdot \mathbf{b} < 0$$

The half usually missing: projection

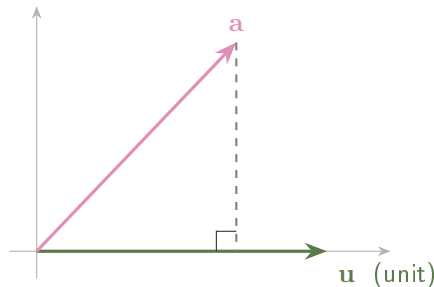


■ Useful fact

$$\mathbf{a} \cdot \mathbf{b} = \underbrace{(\|\mathbf{a}\| \cos \alpha)}_{\text{length of the shadow}} \times \|\mathbf{b}\|, \quad \text{and for unit } \mathbf{u}: \quad \mathbf{a} \cdot \mathbf{u} = \text{signed shadow length.}$$

A coordinate in an orthonormal basis is a dot product.

The half usually missing: projection

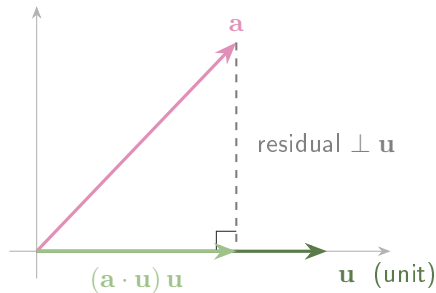


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■ Useful fact

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A coordinate in an orthonormal basis is a dot product.

■ Cauchy—Schwarz

$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$, with equality exactly when $\mathbf{a}$ and $\mathbf{b}$ are dependent.

Divide through: $\cos \alpha \in [-1, 1]$ — the *cosine similarity*.

■ Watch out

The dot product is **not** bounded by 1: $(10, 0) \cdot (10, 0) = 100$.

The bounded quantity is $\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$.

■ Example

A composite score weights the three measures: $\mathbf{w} = (1, 2, -1)$ (reward speed a little, accuracy a lot, penalise anxiety).

Two participants

$$\mathbf{p} = (1, 1, 0) : \quad \mathbf{w} \cdot \mathbf{p} = 1 + 2 - 0 = 3$$

$$\mathbf{q} = (0, 1, 2) : \quad \mathbf{w} \cdot \mathbf{q} = 0 + 2 - 2 = 0$$

$\mathbf{q}$ scores zero — her accuracy is exactly cancelled by her anxiety.

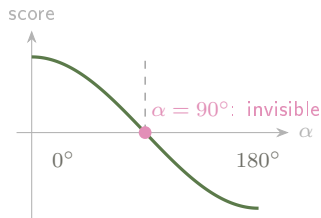
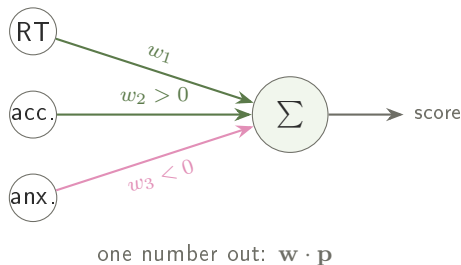
Normalised

$$\|\mathbf{w}\| = \sqrt{1 + 4 + 1} = \sqrt{6} \approx 2.45$$

$$\mathbf{p} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{3}{\sqrt{6}} \approx 1.22$$

$\Rightarrow \mathbf{p}$'s profile *points the same way* as the scoring rule.

How a scoring rule reads a profile



■ Link to cognitive science

Score = $w_1 p_1 + \dots + w_n p_n = \mathbf{w} \cdot \mathbf{p}$. Among profiles of the same size, the score follows $\cos \alpha$: maximal along $\mathbf{w}$, **exactly zero for any profile orthogonal to $\mathbf{w}$** . Such a participant is invisible to this rule — not weakly seen, invisible.

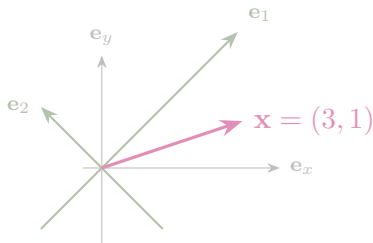
■ Practice — try it

1. Compute $(1, 2, 3) \cdot (4, 0, -1)$ and $\|(3, 4)\|$.
2. Are $(1, 2)$ and $(2, -1)$ orthogonal? What does that mean for a scoring rule with weights $(1, 2)$ applied to the profile $(2, -1)$?
3. Project $\mathbf{a} = (3, 1)$ onto $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$: signed shadow length, then the projection vector.

■ Definition

A **basis**: reference axes in which every vector is written in exactly one way.

Orthonormal: the axes are perpendicular and of length one — and then every coordinate is a projection, $x_i = \mathbf{x} \cdot \mathbf{e}_i$.

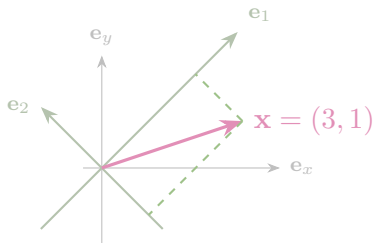


The pink arrow never moves. Grey axes: $(3, 1)$. Tilted axes: $(2\sqrt{2}, -\sqrt{2})$.

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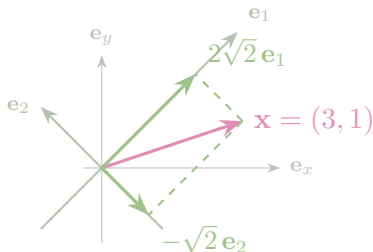


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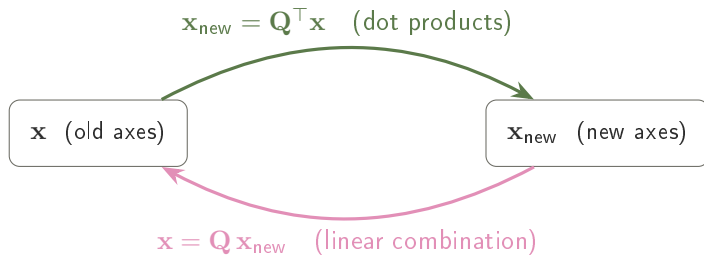
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■ Useful fact

$\mathbf{Q} = (\mathbf{e}_1 \ \mathbf{e}_2)$ holds the new axes as *columns*. Orthonormality is exactly $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$, i.e. $\mathbf{Q}^{-1} = \mathbf{Q}^T$: no inversion is ever needed.

■ Useful fact

1. **Rebuild.** Reconstruct the original vector from its new coordinates.
2. **Compare lengths.** $x_1^2 + x_2^2 = \|\mathbf{x}\|^2$.

The second is Pythagoras — and it is exactly what fails for non-perpendicular axes.

■ Example

Axes rotated by 45° : $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1)$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1, 1)$. Express $\mathbf{x} = (3, 1)$.

$$x_1 = \mathbf{x} \cdot \mathbf{e}_1 = \frac{1}{\sqrt{2}}(3 + 1) = 2\sqrt{2}, \quad x_2 = \mathbf{x} \cdot \mathbf{e}_2 = \frac{1}{\sqrt{2}}(-3 + 1) = -\sqrt{2}.$$

Rebuild

Lengths

$$2\sqrt{2}\mathbf{e}_1 - \sqrt{2}\mathbf{e}_2 = (2, 2) + (1, -1) = (3, 1) \checkmark \quad x_1^2 + x_2^2 = 8 + 2 = 10 = \|\mathbf{x}\|^2 \checkmark$$

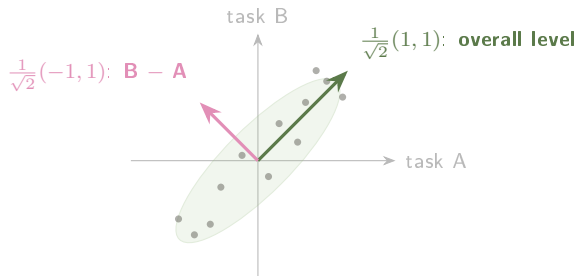
■ Watch out

$$(3\sqrt{2}, -2\sqrt{2}) \quad \text{and} \quad \sqrt{2}(\mathbf{e}_1 - 2\mathbf{e}_2)$$

First: the projections were never taken.

Second: the two coefficients were swapped.

Both are caught instantly by **rebuilding** and by **comparing lengths**.



■ Link to cognitive science

Each dot is a participant's score on two similar tasks (centred). "Task A" and "task B" are the axes the *experiment* handed us; "how good overall" and "which task is easier for them" are the axes the *question* wants. In them the covariance $\begin{pmatrix} v & c \\ c & v \end{pmatrix}$ becomes diagonal — the two new scores are uncorrelated.

■ Practice — try it

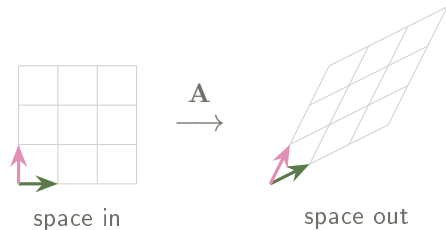
1. Show that $\{\frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1)\}$ is orthonormal.
2. Write $\mathbf{x} = (2, 0)$ in the basis $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1)$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1, 1)$, then rebuild it as a check.
3. Give an orthonormal basis whose first axis points along $(3, 4)$.

A matrix is a machine that moves vectors

■ Definition

$\mathbf{A}$ of size $m \times n$ sends $\mathbf{x} \in \mathbb{R}^n$ to $\mathbf{Ax} \in \mathbb{R}^m$; output slot i is $(\text{row } i) \cdot \mathbf{x}$:

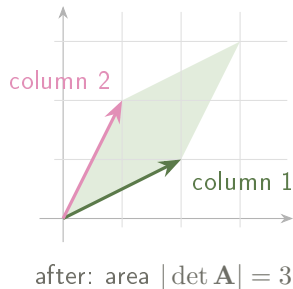
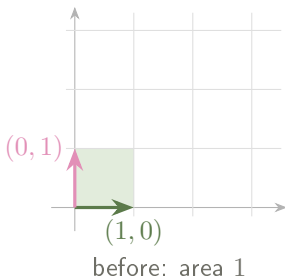
$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{pmatrix}.$$



Rotate, stretch, shear or squash — origin fixed, straight lines straight, grid lines evenly spaced.

A matrix is where the axes land

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$



The columns are where the axis arrows land; everything else follows. This same $\mathbf{A}$ returns in Session 2.

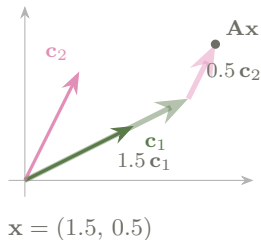
■ Intuition

$$\mathbf{Ax} = x_1 \mathbf{c}_1 + x_2 \mathbf{c}_2$$

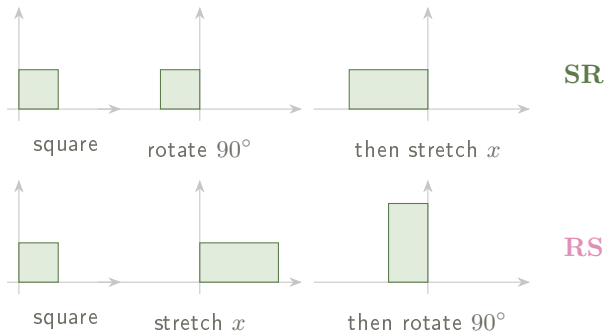
The output is a linear combination of the *columns*.

⇒ the set of possible outputs is exactly $\text{span}(\text{columns})$.

That is what makes *rank* obvious next session.



Order matters: $\mathbf{AB} \neq \mathbf{BA}$



Same two operations, opposite order, different shape. In $\mathbf{AB}$ the *right-hand* matrix acts first; $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ reverses too.

■ Example

Recode two task scores as *mean* and *difference*: $\mathbf{M} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix}$, scores $\mathbf{s} = (8, 4)$.

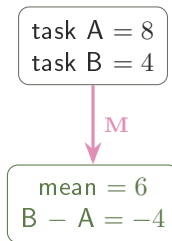
By rows

$$\mathbf{Ms} = \begin{pmatrix} (\frac{1}{2}, \frac{1}{2}) \cdot (8, 4) \\ (-1, 1) \cdot (8, 4) \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

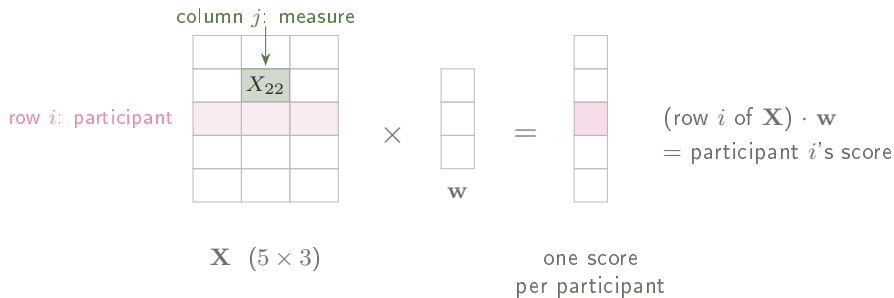
By columns

$$8 \begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix} + 4 \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} \checkmark$$

Same participant, two numbers in, two numbers out — and the second pair is the one you would actually analyse.



The data matrix



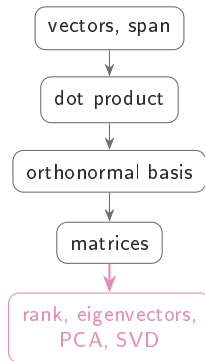
■ Link to cognitive science

X_{ij} : **participant i , measure j** — rows are people, columns are variables. Then $\mathbf{X}\mathbf{w}$ applies one scoring rule to *everybody at once*, in a single product. Predicting an outcome from several measures is exactly this: the linear part of a regression is $\mathbf{X}\mathbf{w}$.

■ Practice — try it

1. Compute $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$; say in words what this matrix does.
2. Compute $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 5 \\ -2 \end{pmatrix}$. Which geometric operation is it?
3. Compute $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ and the same product the other way. Comment.

- **Span** = what a set of vectors can build; **independence** = no redundancy.
- **Dot product** $\Rightarrow$ length, angle, and the **projection** picture.
- **Change of basis**: out with $\mathbf{Q}^T$, back with $\mathbf{Q}$ — checked twice.
- **Matrix** = transformation: rows to compute, columns to understand.



Next session: what a transformation does to space — and how many dimensions a data table really has.